Year 13 Maths - General binomial expansion

Similar questions - Binomial expansion for any power

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Find the first four terms in the expansion of $(1+4x)^{-1/2}$ in ascending powers of x, and state the range of values of x for which the expansion is valid. **[5]**

(b) Find the coefficient of x^2 in the expansion of $\dfrac{2+x}{\sqrt{1+4x}}$. **[3]**

(c) Use your expansion from part **(a)** with $x = 0.01$ to estimate $\dfrac{1}{\sqrt{1.04}}$ to 5 decimal places, and comment on the accuracy of your estimate. **[3]**

2

(a) Find the first three terms in the expansion of $(9-2x)^{3/2}$ in ascending powers of x, and state the range of values of x for which the expansion is valid. **[5]**

(b) In the expansion of $(9-2x)^{3/2}(1+ax)^2$ the coefficient of x^2 is $\frac{13}{2}$. Find the possible values of the constant a, giving your answers in exact form. **[4]**

Answers

Question 1

(a) $(1+u)^{-1/2} = 1 - \frac{1}{2}u + \frac{3}{8}u^2 - \frac{5}{16}u^3$ with $u = 4x$: $1 - 2x + 6x^2 - 20x^3$. Valid for $|4x| < 1$, i.e. $|x| < \frac{1}{4}$

(b) $(2+x)(1-2x+6x^2+\ldots)$: coefficient of x^2 is $2(6)+1(-2) = 10$

(c) $1 - 0.02 + 0.0006 - 0.00002 = 0.98058$. The calculator value of $1/\sqrt{1.04}$ is 0.98058, so the estimate is accurate to 5 decimal places ($x = 0.01$ is well within $|x| < \frac{1}{4}$)

Question 2

(a) $(9-2x)^{3/2} = 9^{3/2}\left(1 - \frac{2x}{9}\right)^{3/2} = 27\left[1 + \frac{3}{2}\left(-\frac{2x}{9}\right) + \frac{(3/2)(1/2)}{2}\left(\frac{2x}{9}\right)^2\right] = 27 - 9x + \frac{1}{2}x^2$. Valid for $\left|\frac{2x}{9}\right| < 1$, i.e. $|x| < \frac{9}{2}$

(b) $(1+ax)^2 = 1 + 2ax + a^2x^2$. Coefficient of x^2: $\frac{1}{2} + (-9)(2a) + 27a^2 = \frac{13}{2}$, so $27a^2 - 18a - 6 = 0$, i.e. $9a^2 - 6a - 2 = 0$, giving $a = \dfrac{6 \pm \sqrt{108}}{18} = \dfrac{1 \pm \sqrt{3}}{3}$